MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 03** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 10: Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

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Q1. If angle between two radii of a circle is 130°, then the angle between the tangents at the ends of these radii is: [1]

(A) 90° (B) 50° (C) 70° (D) 40°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 50°
Justification: Tangents and radii form a cyclic quadrilateral with two 90° angles, so angle between tangents = $180° - 130° = \boxed{\mathbf{50°}}$.

Q2. The pair of tangents AP and AQ drawn from an external point A to a circle with centre O are perpendicular to each other and length of each tangent is 5 cm. The radius of the circle is: [1]

(A) 10 cm (B) 7.5 cm (C) 5 cm (D) 2.5 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (C) 5 cm
Justification: With $\angle PAQ = 90°$, $\angle OPA = \angle OQA = 90°$, and $OP = OQ$, quadrilateral $OPAQ$ is a square. Hence radius $OP = AP = \boxed{\mathbf{5}\text{ cm}}$.

Q3. Two concentric circles have radii 13 cm and 5 cm. The length of the chord of the larger circle which touches the smaller circle is: [1]

(A) 12 cm (B) 24 cm (C) 18 cm (D) 20 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 24 cm
Justification: Half-chord length $= \sqrt{13^2 - 5^2} = \sqrt{144} = 12$ cm $\implies$ Chord $= 2 \times 12 = \boxed{\mathbf{24}\text{ cm}}$.

Q4. If tangents PA and PB from a point P to a circle with centre O are inclined at $60°$ and $OP = 10$ cm, then length of tangent PA is: [1]

(A) 5 cm (B) $5\sqrt{3}$ cm (C) $10\sqrt{3}$ cm (D) $5/\sqrt{3}$ cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $5\sqrt{3}$ cm

Justification: OP bisects $\angle APB$, so $\angle OPA = 30°$. In right $\triangle OAP$: $\cos 30° = \frac{PA}{OP} \implies PA = 10 \times \frac{\sqrt{3}}{2} = \boxed{\mathbf{5\sqrt{3}}\text{ cm}}$.

Q5. **Assertion (A):** A parallelogram circumscribing a circle is always a rhombus.
Reason (R): In any quadrilateral circumscribing a circle, the sum of opposite sides is equal, so $AB + CD = AD + BC$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: By (R), $AB + CD = AD + BC$. Since opposite sides of a parallelogram are equal ($AB = CD$ and $AD = BC$), we get $2AB = 2AD \implies AB = AD$, making all four sides equal (a rhombus). Thus (A) and (R) are true and (R) explains (A).

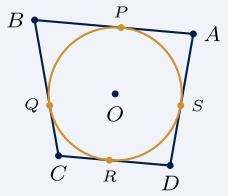
SECTION B: Very Short Answer Type

MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let the circle touch AB, BC, CD, DA at P, Q, R, S respectively.



Step 1: Equal Tangents from Vertices [1 Mark]
By Theorem 10.2, tangents drawn from an external point to a circle are equal:
$AP = AS, \quad BP = BQ, \quad CR = CQ, \quad DR = DS.$

Step 2: Adding Opposite Segments [1 Mark]
$(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$

$$\boxed{\therefore AB + CD = AD + BC \quad \text{[Proved]}}$$

Q7. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

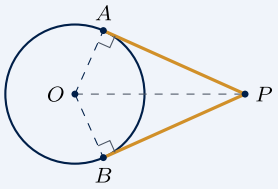
Given: Circle centred at O; external point P; tangents PA, PB touching at A, B.

To Prove: $\angle APB + \angle AOB = 180°$. [0.5 Mark]

Proof & Deduction: [2.5 Marks]

1. Since radius $\perp$ tangent at point of contact (Theorem 10.1): $OA \perp PA \implies \angle OAP = 90°$ and $OB \perp PB \implies \angle OBP = 90°$.
2. By angle sum property in quadrilateral $OAPB$:
$\angle OAP + \angle APB + \angle OBP + \angle AOB = 360°$
$90° + \angle APB + 90° + \angle AOB = 360°$

$$\boxed{\therefore \angle APB + \angle AOB = 180° \quad \text{[Proved]}}$$



Q8. PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q intersect at point T. Find the length of tangent TP. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Join OT intersecting chord PQ at M. OT is the perpendicular bisector of PQ.

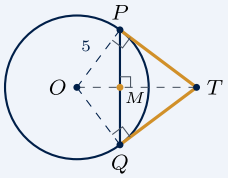
Step 1: Finding OM [1 Mark]

$PM = \frac{PQ}{2} = 4$ cm. In right $\triangle OPM$: $OM = \sqrt{OP^2 - PM^2} = \sqrt{5^2 - 4^2} = 3$ cm.

Step 2: Similarity $\triangle OPT \sim \triangle PMO$ [2 Marks]

$\angle OPT = \angle PMO = 90°$ and $\angle POT = \angle MOP$ (common). By AA similarity, $\triangle OPT \sim \triangle PMO$:

$$\frac{TP}{PM} = \frac{OP}{OM} \implies \frac{TP}{4} = \frac{5}{3} \implies \boxed{TP = \mathbf{\frac{20}{3}} \text{ cm} \quad \left(\text{or } \mathbf{6\frac{2}{3}} \text{ cm}\right)}$$



Q9. A quadrilateral $ABCD$ circumscribes a circle with centre O. Prove that opposite sides subtend supplementary angles at the centre, i.e., $\angle AOB + \angle COD = 180°$. [3]

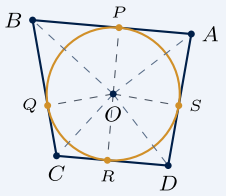
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Construction: Let sides touch circle at P, Q, R, S. Join OA, OB, OC, OD and radii OP, OQ, OR, OS. [0.5 Mark]

Proof & Deduction: [2.5 Marks]

1. In $\triangle OAP$ and $\triangle OAS$: $OA = OA$ (common), $OP = OS$ (radii), $AP = AS$ (tangents). By SSS congruence, $\triangle OAP \cong \triangle OAS \implies \angle AOP = \angle AOS$.
2. Similarly, $\angle BOP = \angle BOQ$, $\angle COQ = \angle COR$, and $\angle DOR = \angle DOS$.
3. Sum around centre O: $2(\angle AOP + \angle BOP + \angle COR + \angle DOR) = 360° \implies (\angle AOP + \angle BOP) + (\angle COR + \angle DOR) = 180°$.

$$\boxed{\therefore \angle AOB + \angle COD = 180° \quad \text{[Proved]}}$$



Q10. Highway Traffic Roundabout Slip-Lane Design

Civil engineers design a circular traffic roundabout of radius $r = 15$ m centred at O. Two straight slip lanes PA and PB meet at an intersection point P outside the roundabout and touch the circular island at points A and B. The angle between the slip lanes is $\angle APB = 60°$.

(i) Find the measure of the central angle $\angle AOB$ subtended by the slip-lane entry points. [1]

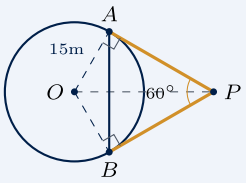
(ii) Calculate the straight-line distance OP from the intersection to the roundabout centre. [1]

(iii) Calculate the length of each slip lane PA and PB, OR prove that $\triangle PAB$ forms an equilateral triangle. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Central Angle $\angle AOB$: [1 Mark]

$\angle AOB = 180° - \angle APB = 180° - 60° = \boxed{\mathbf{120°}}$.



(ii) Straight-line Distance OP: [1 Mark]

OP bisects $\angle APB$, so $\angle OPA = 30°$. In right $\triangle OAP$: $\sin 30° = \frac{OA}{OP} \implies \frac{1}{2} = \frac{15}{OP} \implies \boxed{OP = \mathbf{30}\text{ m}}$.

(iii) Main Choice — Length of Each Slip Lane (PA and PB): [2 Marks]

In right $\triangle OAP$: $\tan 30° = \frac{OA}{PA} \implies \frac{1}{\sqrt{3}} = \frac{15}{PA} \implies \boxed{PA = PB = \mathbf{15\sqrt{3}}\text{ m} \quad (\approx \mathbf{25.98}\text{ m})}$.

— OR — (Alternative Choice — Proof that $\triangle PAB$ is Equilateral): [2 Marks]

Since $PA = PB$ (Theorem 10.2), $\triangle PAB$ is isosceles with $\angle PAB = \angle PBA = \frac{180° - 60°}{2} = 60°$. As all three interior angles equal 60°, $\boxed{\mathbf{\triangle PAB}\text{ is an equilateral triangle [Proved]}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★